

SOLUTION We wish to use the formula $\int \cos u \, du = \sin u + C$, so let us make the substitution

$$u = \sqrt{x} = x^{1/2}, \quad du = \frac{1}{2} x^{-1/2} dx = \frac{1}{2\sqrt{x}} dx.$$

If we introduce the factor $\frac{1}{2}$ into the integrand and compensate by multiplying the integral by 2, we obtain

$$\begin{aligned} \int \frac{\cos \sqrt{x}}{\sqrt{x}} dx &= 2 \int \cos \sqrt{x} \left(\frac{1}{2} \cdot \frac{1}{\sqrt{x}} \right) dx \\ &= 2 \int \cos u \, du = 2 \sin u + C \\ &= 2 \sin \sqrt{x} + C. \end{aligned}$$

EXAMPLE 7 Evaluate $\int \cos^3 5x \sin 5x \, dx$.

SOLUTION The form of the integrand suggests that we use the power rule $\int u^3 \, du = \frac{1}{4}u^4 + C$. Thus, we let

$$u = \cos 5x, \quad du = -5 \sin 5x \, dx.$$

The form of du indicates that we should introduce the factor -5 into the integrand, multiply the integral by $-\frac{1}{5}$, and then integrate as follows:

$$\begin{aligned} \int \cos^3 5x \sin 5x \, dx &= -\frac{1}{5} \int \cos^3 5x (-5 \sin 5x) \, dx \\ &= -\frac{1}{5} \int u^3 \, du \\ &= -\frac{1}{5} \left(\frac{u^4}{4} \right) + C \\ &= -\frac{1}{20} \cos^4 5x + C \end{aligned}$$

EXERCISES 5.2

Exer. 1–8: Evaluate the integral using the given substitution, and express the answer in terms of x .

1 $\int x(2x^2 + 3)^{10} \, dx; \quad u = 2x^2 + 3$

2 $\int \frac{x}{(x^2 + 5)^3} \, dx; \quad u = x^2 + 5$

3 $\int x^2 \sqrt[3]{3x^3 + 7} \, dx; \quad u = 3x^3 + 7$

4 $\int \frac{5x}{\sqrt{x^2 - 3}} \, dx; \quad u = x^2 - 3$

5 $\int \frac{(1 + \sqrt{x})^3}{\sqrt{x}} \, dx; \quad u = 1 + \sqrt{x}$

6 $\int \frac{1}{(5x - 4)^{10}} \, dx; \quad u = 5x - 4$

7 $\int \sqrt{x} \cos \sqrt{x^3} \, dx; \quad u = x^{3/2}$

8 $\int \tan x \sec^2 x \, dx; \quad u = \tan x$

Exer. 9–48: Evaluate the integral.

9 $\int \sqrt{3x - 2} \, dx$

10 $\int \sqrt[4]{2x + 5} \, dx$

11 $\int \sqrt[3]{8t + 5} \, dt$

12 $\int \frac{1}{\sqrt{4 - 5t}} \, dt$

13 $\int (3z + 1)^4 \, dz$

14 $\int (2z^2 - 3)^5 z \, dz$

15 $\int v^2 \sqrt{v^3 - 1} \, dv$

16 $\int v \sqrt{9 - v^2} \, dv$

17 $\int \frac{x}{\sqrt[3]{1 - 2x^2}} \, dx$

18 $\int (3 - x^4)^3 x^3 \, dx$

- 19 $\int (s^2 + 1)^2 ds$ 20 $\int (3 - s^3)^2 s ds$
- 21 $\int \frac{(\sqrt{x} + 3)^4}{\sqrt{x}} dx$ 22 $\int \left(1 + \frac{1}{x}\right)^{-3} \left(\frac{1}{x^2}\right) dx$
- 23 $\int \frac{t - 2}{(t^2 - 4t + 3)^3} dt$ 24 $\int \frac{t^2 + t}{(4 - 3t^2 - 2t^3)^4} dt$
- 25 $\int 3 \sin 4x dx$ 26 $\int 4 \cos \frac{1}{2}x dx$
- 27 $\int \cos (4x - 3) dx$ 28 $\int \sin (1 + 6x) dx$
- 29 $\int v \sin (v^2) dv$ 30 $\int \frac{\cos \sqrt[3]{v}}{\sqrt[3]{v^2}} dv$
- 31 $\int \cos 3x \sqrt[3]{\sin 3x} dx$ 32 $\int \frac{\sin 2x}{\sqrt{1 - \cos 2x}} dx$
- 33 $\int (\sin x + \cos x)^2 dx$ (Hint: $\sin 2\theta = 2 \sin \theta \cos \theta$)
- 34 $\int \frac{\sin 4x}{\cos 2x} dx$ (Hint: $\sin 2\theta = 2 \sin \theta \cos \theta$)
- 35 $\int \sin x (1 + \cos x)^2 dx$ 36 $\int \sin^3 x \cos x dx$
- 37 $\int \frac{\sin x}{\cos^4 x} dx$ 38 $\int \sin 2x \sec^5 2x dx$
- 39 $\int \frac{\cos t}{(1 - \sin t)^2} dt$ 40 $\int (2 + 5 \cos t)^3 \sin t dt$
- 41 $\int \sec^2 (3x - 4) dx$ 42 $\int \frac{\csc 2x}{\sin 2x} dx$
- 43 $\int \sec^2 3x \tan 3x dx$ 44 $\int \frac{1}{\tan 4x \sin 4x} dx$
- 45 $\int \frac{1}{\sin^2 5x} dx$ 46 $\int \frac{x}{\cos^2 (x^2)} dx$
- 47 $\int x \cot (x^2) \csc (x^2) dx$ 48 $\int \sec \left(\frac{x}{3}\right) \tan \left(\frac{x}{3}\right) dx$

Exer. 49–52: Solve the differential equation subject to the given conditions.

- 49 $f'(x) = \sqrt[3]{3x + 2};$ $f(2) = 9$
- 50 $\frac{dy}{dx} = x\sqrt{x^2 + 5};$ $y = 12$ if $x = 2$
- 51 $f''(x) = 16 \cos 2x - 3 \sin x;$ $f(0) = -2; \quad f'(0) = 4$
- 52 $f''(x) = 4 \sin 2x + 16 \cos 4x;$ $f(0) = 6; \quad f'(0) = 1$

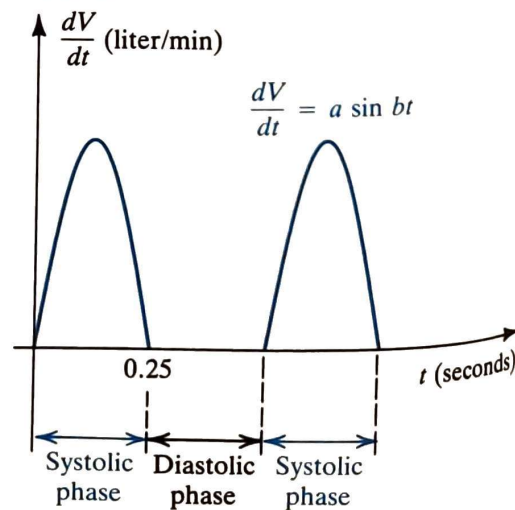
Exer. 53–56: Evaluate the integral by (a) the method of substitution and (b) expanding the integrand. In what way do the constants of integration differ?

- 53 $\int (x + 4)^2 dx$ 54 $\int (x^2 + 4)^2 x dx$

55 $\int \frac{(\sqrt{x} + 3)^2}{\sqrt{x}} dx$ 56 $\int \left(1 + \frac{1}{x}\right)^2 \frac{1}{x^2} dx$

- 57 A charged particle is moving on a coordinate line in a magnetic field such that its velocity (in cm/sec) at time t is given by $v(t) = \frac{1}{2} \sin(3t - \frac{1}{4}\pi)$. Show that the motion is simple harmonic (see page 223).
- 58 The acceleration of a particle that is moving on a coordinate line is given by $a(t) = k \cos(\omega t + \phi)$ for constants k , ω , and ϕ and time t (in seconds). Show that the motion is simple harmonic (see page 223).
- 59 A reservoir supplies water to a community. In summer the demand A for water (in ft^3/day) changes according to the formula $dA/dt = 4000 + 2000 \sin(\frac{1}{90}\pi t)$ for time t (in days), with $t = 0$ corresponding to the beginning of summer. Estimate the total water consumption during 90 days of summer.
- 60 The pumping action of the heart consists of the systolic phase, in which blood rushes from the left ventricle into the aorta, and the diastolic phase, during which the heart muscle relaxes. The graph shown in the figure is sometimes used to model one complete cycle of the process. For a particular individual, the systolic phase lasts $\frac{1}{4}$ second and has a maximum flow rate dV/dt of 8 L/min, where V is the volume of blood in the heart at time t .
- (a) Show that $dV/dt = 8 \sin(240\pi t)$ L/min.
- (b) Estimate the total amount of blood pumped into the aorta during a systolic phase.

EXERCISE 60



- 61 The rhythmic process of breathing consists of alternating periods of inhaling and exhaling. For an adult, one complete cycle normally takes place every 5 seconds. If V denotes the volume of air in the lungs at time t , then dV/dt is the flow rate.